



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY
IV SEMESTER B.TECH. MID SEMESTER EXAMINATION
MAT 2238

Date of Examination: 08/03/2025 Time: 08:30 AM-10:00AM MAX.MARKS: 30

VECTOR ANALYSIS AND COMPLEX VARIABLES [MAT 2238]

Marks: 30

Duration: 90 mins.

MCQ

Answer all the questions.

Section Duration: 20 mins

Please select the correct answer from the options given

1) Which of the following vector fields is not irrotational?

1) $\vec{F} = (yz, xz, xy)$	2) $\vec{F} = (x, y, z)$	3) $\vec{F} = (-y, x, 0)$	4) $\vec{F} = (x^2, y^2, z^2)$
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Correct option is: 3

2) The directional derivative of a scalar valued function f at a point (x_0, y_0) in the direction of a non-zero (constant) vector $\vec{b} = (b_1, b_2)$ is denoted by $D_{\vec{b}}(f)(x_0, y_0)$. Which of the following is true?

1) $D_{(1,1)}(f) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$	2) $D_{(2,0)}(f) = \frac{\partial f}{\partial x} = D_{(1,0)}(f)$	3) $D_{(0,2)}(f) = 2 \frac{\partial f}{\partial y}$	4) $D_{(0,1)}(f) = \frac{\partial f}{\partial x}$
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Correct option is: 2

3) Let $\vec{F}(x, y, z) = (2xz, -y^2, 3xy)$ be a vector field. Then, the divergence of $\vec{F}$ at the point $(1, 2, -1)$ is

1) -10	2) -6	3) 2	4) 10
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(1)

Correct option is: 2

4) $\lim_{z \rightarrow z_0} f(z) = L$
Assume that the limit $\lim_{z \rightarrow z_0} f(z)$ exists, where f is a complex valued function of a complex variable. Which of the following statements is its equivalent?

1) For every $\delta > 0$, there exists an $\epsilon > 0$ such that if $ z - z_0 < \delta$, then $ f(z) - L < \epsilon$.	2) For every $\epsilon > 0$, there exists a $\delta > 0$ such that if $0 < z - z_0 < \delta$, then $ f(z) - L < \epsilon$.	3) For every $\epsilon > 0$, there exists a $\delta > 0$ such that if $0 < z - z_0 < \delta$, then $ f(z) - L < \epsilon$.	4) For every $\delta > 0$, there exists an $\epsilon > 0$ such that if $0 < f(z) - L < \epsilon$, then $ z - z_0 < \delta$.
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(1)

Correct option is: 3

5) Let $f(z) = u(x, y) + iv(x, y)$ be a complex valued function defined on a (non-empty)

(1)

domain $D \subseteq \mathbb{C}$. Let $z_0 = x_0 + iy_0 \in D$. Consider the following statements.

I) f is complex differentiable at z_0 .

II) u_x, u_y, v_x and v_y exist at (x_0, y_0) and satisfy $u_x = v_y, u_y = -v_x$ at (x_0, y_0) .

III) u_x, u_y, v_x and v_y exist in a neighbourhood of (x_0, y_0) and satisfy $u_x = v_y, v_x = -u_y$ at (x_0, y_0) .

IV) u_x, u_y, v_x and v_y exist in a neighbourhood of (x_0, y_0) , are continuous at (x_0, y_0) and satisfy $v_x = -u_y, v_y = u_x$ at (x_0, y_0) .

V) u_x, u_y, v_x and v_y are continuous on a neighbourhood of (x_0, y_0) and satisfy $u_x = v_y, u_y = -v_x$ at (x_0, y_0) .

Which of the following is/are true?

1)	Each statement from II to V individually imply statement I.	2)	$V \Rightarrow IV$ but $IV \not\Rightarrow I$	3)	$V \Rightarrow I$ but $IV \not\Rightarrow I$	4)	Both IV and V individually imply I.
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Correct option is: 4

Descriptive

Answer all the questions.

Answer all the questions. Any missing data can be assumed suitably with proper reasoning.

6) Let f be a scalar field having continuous first order partial derivatives everywhere in $\mathbb{R}^3$. Let c be a real constant and let $L(c)$ denote the corresponding level surface of f . For $P \in L(c)$, if $\nabla f(P) \neq \vec{0}$, show that it is a surface normal to $L(c)$ at P . (3)

7) Find an equation for the tangent plane to the surface $xz^2 + x^2y = z - 1$ at the point $(1, -3, 2)$. (3)

8) Find the angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point $(1, -2, 1)$. (3)

9) Let $\vec{v}(x, y, z)$ be a continuously differentiable vector field that is also conservative. Then, determine $\text{curl}(\vec{v})$. (3)

10) Using the $\epsilon - \delta$ definition of limit, show that for $z = x + iy$,

$$\lim_{z \rightarrow 1-i} x + i(2x + y) = 1 + i$$
 (3)

11) Discuss the limiting behaviour of the complex valued function $f(z) = \frac{z}{\bar{z}}$, $z \neq 0$, (2)

as z approaches 0? What happens if z approaches 1? Justify your answer with sufficient explanation.

12) Consider the function $f(z) = |z|^2$.

- 1) Determine all the points (in the complex plane) where it satisfies Cauchy-Riemann equations.
 - 2) Determine all the points where it is differentiable.
 - 3) Determine all the points where it is analytic.
- (2)

Write the computations and justifications clearly.

13) Let particle A of mass M be fixed at a point P_0 and let particle B of mass m be free to take up various positions P in space. Particle A is known to attract B with a gravitational force (directed from P to P_0) given by

$$\vec{p} = \frac{-GMm}{r^3} \vec{r}. \quad (4)$$

Show that $\vec{p}$ is a conservative vector field. Here, G is the universal gravitational constant.

14) Find all the four complex roots of $-8 - 8\sqrt{3}i$ in rectangular coordinates. (2)

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